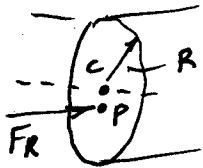


FLUID PRESSURE: F_R for FLAT PLATES OF VARIABLE WIDTH

A SIMPLE FORMULA, EASIER THAN INTEGRATION!

Esp. helpful for F_R at end of a tank:
Half circle



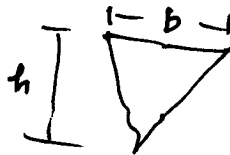
$$c = \bar{z} = R$$

$$A = \pi R^2$$



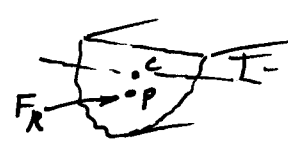
$$c = \bar{z} = \frac{h}{3}$$

$$A = \frac{1}{2}bh$$



$$c = \frac{2}{5}h = \bar{z} \text{ for parabola}$$

$$A = 2\left(\frac{2}{3}bh\right)$$



$$c = \bar{z} = \frac{4R}{3\pi}$$

$$A = \frac{\pi}{2}R^2$$



All of these are FLAT plates of variable width.
 F_R acts at "P" = Center of Pressure at a depth slightly deeper than the centroid of the plate, C.

Simple formula for F_R :

$$F_R = (P \text{ at } \bar{z}) * (\text{Area of Plate})$$

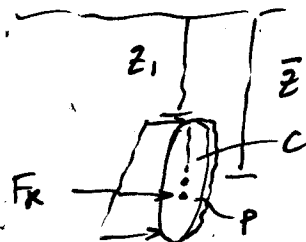
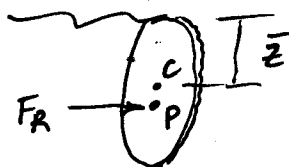
or

$$F_R = \gamma \cdot \bar{z} \cdot A \sim \text{acts at Center of Pressure, P.}$$

The plate can be at any depth:

$\bar{z}$ from free water surface...

$$\bar{z} = z_1 + c$$



Derivation of this formula

General submerged flat plate.

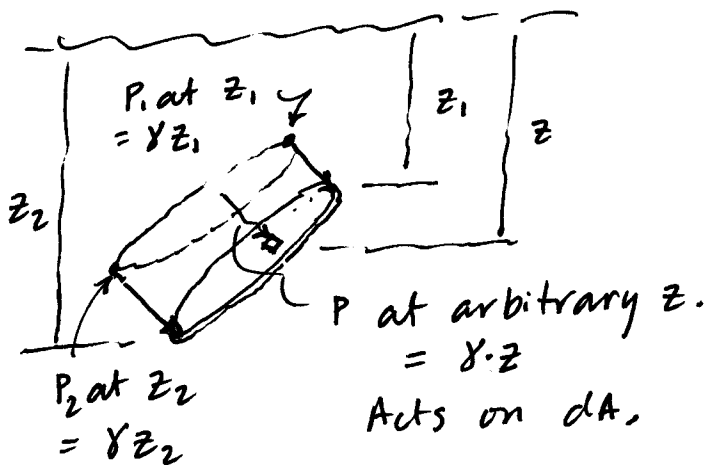
There is a pressure distribution acting on the plate...

P_s act $\perp$ to plate

$$P_s = f(\text{depth})$$

$$\text{At top, } P_1 = \gamma z_1$$

$$\text{At bottom, } P_2 = \gamma z_2$$



In general,
$$F_R = \text{VOLUME OF THIS PRESSURE DISTRIB} = \int_A P dA$$

but $p = \gamma \cdot z$, so
$$F_R = \int_A \gamma \cdot z \cdot dA$$

$$F_R = \gamma \cdot \int_A z dA$$

But this $\int_A z dA$ integral looks familiar!

Remember the centroid definition:

$$\bar{z} = \frac{\int_A z dA}{\int_A dA}$$

Numerator of centroid definition!

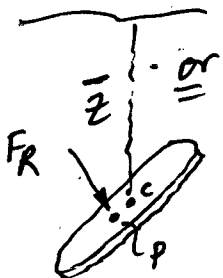
thus
$$\int_A z dA = \bar{z} \cdot \int_A dA = \bar{z} \cdot A$$

Finally,
$$F_R = \gamma \cdot \bar{z} \cdot A$$

Area, A

$$F_R = (P \text{ at depth } \bar{z} \text{ to centroid of plate}) \cdot A$$

Important! Though F_R is calculated from p at $\bar{z}$, F_R does not act at c !!
 F_R acts at the "center of pressure", P .

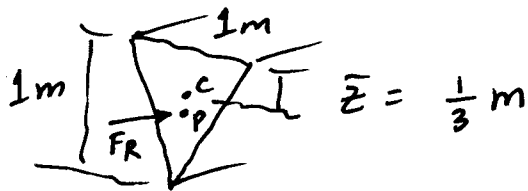


Applying this formula to problems we've previously integrated for F_R .

① Hibbeler Ex 9-16

$$\rho_w = 1000 \text{ kg/m}^3$$

$$\gamma = 9810 \text{ N/m}^3$$



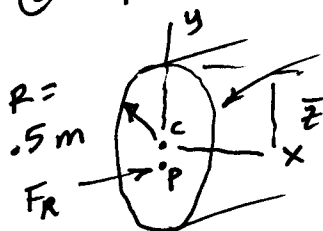
$$F_R = (\bar{p} \text{ at } \bar{z}) \cdot \text{Area}$$

$$= \gamma \cdot \bar{z} \cdot A$$

$$F_R = (\underbrace{9810}_{\gamma}) (\underbrace{\frac{1}{3} \text{ m}}_{\bar{z}}) \cdot \left[\underbrace{\frac{1}{2} (1) (1)}_{\frac{1}{2} b h} \right]$$

$$F_R = 1.635 \text{ kN} ! \quad \sim \text{compare to integration!}$$

② Round Tank from old exam



$$x^2 + y^2 = .5^2$$

$$F_R = \gamma \cdot \bar{z} \cdot A$$

$$\rho = 1 \text{ Mg/m}^3 = 1000 \text{ kg/m}^3$$

$$\gamma = 9810 \text{ N/m}^3$$

$$\bar{z} = .5 \text{ m}$$

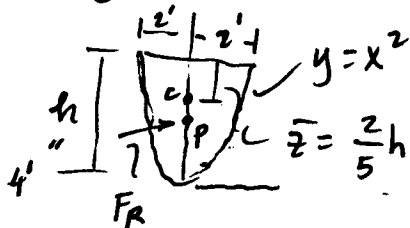
$$A = \pi (.5)^2$$

$$F_R = (9810) (.5) (\pi (.5)^2) = \boxed{3.85 \text{ kN} = F_R}$$

③ HW 9-126

Parabola

$$\gamma = 25 \text{ lb/ft}^3 \quad \left. \vphantom{\gamma = 25 \text{ lb/ft}^3} \right\} \text{Very light fluid!!}$$



$$\bar{z} = \frac{2}{5} (4) = .4 (4) = 1.6 \text{ ft}$$

$$A = 2 \left[\frac{2}{3} b h \right] = 2 \left[\frac{2}{3} (2) (4) \right]$$

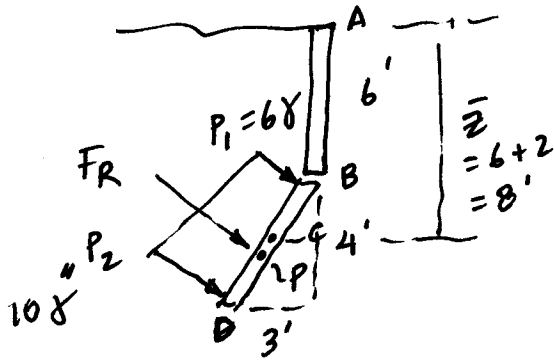
$$= 2 (16/3) = 32/3 \text{ ft}^2$$

$$F_R = \gamma \cdot \bar{z} \cdot A = (25 \frac{\text{lb}}{\text{ft}^3}) (1.6 \text{ ft}) (\frac{32}{3} \text{ ft}^2)$$

$$F_R = 426.7 \text{ lb}$$

Does this apply to a square plate?
Should we use it for square plates?

Plate is 1 ft deep into page.



$$F_R = \left(\frac{6\gamma + 10\gamma}{2} \right) (L) (1)$$

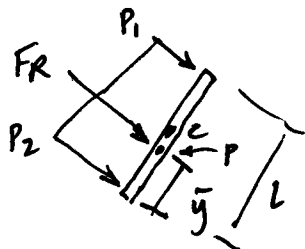
" 5 ft

$$= (8\gamma) (5) (1)$$

$$F_R = 40\gamma$$

$$\gamma = 62.4 \frac{\text{lb}}{\text{ft}^3}$$

$$F_R = 2.50 \text{ kip}$$



$$\bar{y} = \frac{1}{3} \left[\frac{2P_1 + P_2}{P_1 + P_2} \right] L$$

Not at
center of plate, C.

Using the $F_R = \gamma \cdot \bar{z} \cdot A$

formula ...

$$\gamma = 62.4 \text{ lb/ft}^3$$

$$\bar{z} = 8'$$

$$A = L(1) = 5(1)$$

Thus, $F_R = (62.4) (8 \text{ ft}) (5 \cdot 1)$

γ $\bar{z}$ A

$$F_R = 40\gamma$$

$$F_R = 2.50 \text{ kip}$$

Conclusion:
Can use either
approach, but
 F_R acts at $\bar{y}$,
at center of pressure,
NOT at $\bar{z}$.

Acts at $\bar{y}$ (not at C!): $\bar{y} = \frac{1}{3} \left[\frac{2(6\gamma) + 10\gamma}{6\gamma + 10\gamma} \right] (5)$

$$\bar{y} = \frac{5}{3} \left(\frac{22}{16} \right) = 2.292 \text{ ft} = \bar{y}$$

measured from D
at P_2 .